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THE MULTILINEAR GAUGE-INVARIANT OPERATOR BASIS FOR FIVE MASSLESS PHOTONS: REPRESENTATION THEORY, INVARIANT RING, AND A COMPLETE LOW-DEGREE CLASSIFICATION

MARTÍN BACA

ABSTRACT. We construct the S_5 -covariant kinematic data (Mandelstam invariants and photon polarization data) for tree-level five-photon scattering in flat space, in generic spacetime dimension D (no Gram-determinant degeneracies). We show that the space of independent Mandelstam invariants for five massless momenta carries an *irreducible* 5-dimensional representation of S_5 , genuinely richer than the $n = 4$ case, where S_3 acts by simple permutation of (s, t, u) . We compute the complete invariant ring of this representation (5 primary invariants of degrees 2–6, 6 secondary invariants of degrees 0, 6, 7, 8, 9, 15) and cross-validate it exactly against Henning, Lu, Melia & Murayama [1], who solved the corresponding *scalar* $n = 5$ problem – confirming our representation-theoretic machinery before extending it to spin. We then derive the gauge-invariant polarization data structure for photons at $n = 5$: each particle carries *two* independent physical parameters (not one, as at $n = 4$), with closed-form expressions for them and their inverse. We identify four families of manifestly gauge-invariant building blocks (pairwise field-strength contractions, a linear “singlet”, a double field-strength “sandwich”, and – new – a *triple* field-strength sandwich), prove a parity theorem forcing all multilinear five-photon invariants to have odd total momentum degree, and give an exact classification of the space of such invariants through mass dimension 11: dimension 0 at degree 7, 0 at degrees 8 and 10 (parity), 3 at degree 9, and 16 at degree 11 (15 from previously known building blocks plus 1 new dimension from the triple sandwich). All results are cross-checked by at least two independent methods, and several real errors caught and corrected during verification are documented for transparency.

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1. INTRODUCTION

Classifying the tree-level S -matrices of massless particles in flat space, order by order in derivatives, is a long-standing question in the amplitudes/EFT literature. For scalars, Henning, Lu, Melia & Murayama (HLMM) [1] gave a systematic method: the independent kinematic invariants for n massless scalar legs carry a representation of the permutation group S_n , and the space of allowed operators at each mass dimension is the Hilbert series of the corresponding invariant ring. Chowdhury, Gadde, Gopak, Halder, Janagal & Minwalla (CGGHJM) [2] specialized this program to $n = 4$ particles *with spin* (photons and gravitons), giving a complete, unconditional classification and parameterization of all local four-photon and four-graviton polynomial S -matrices at any derivative order; they then additionally imposed the *Classical Regge Growth* (CRG) conjecture — that any consistent classical S -matrix grows no faster than s^2 in the Regge limit — as a further, conjectural physical criterion to select the subclass of these already-classified operators that is CRG-compatible (e.g. isolating a four-parameter family for photons, and ruling out any $D \leq 6$ polynomial correction to the Einstein graviton amplitude).

The natural next step, $n = 5$, splits into two problems of very different difficulty:

- The **scalar** case was solved completely by HLMM themselves [1], §5.6.1: the relevant invariant ring, its primary/secondary invariant degrees, and Hilbert series are all known.
- The **spinning** case is left open by HLMM, who note (§5.4.4, in the context of their discussion of the operator basis at $n > 4$) that rings for spinning operators are, in general, not Cohen–Macaulay, unlike the scalar rings (which they prove are Cohen–Macaulay for all n). This underlies the *graviton* case of open question #44 of the “100 Open Questions” survey of Strings 2024 [3], which asks whether the tree-level Einstein S -matrix is the unique consistent classical n -graviton S -matrix (with no high-spin exchange poles), noting this is established only for $n = 4$ (via CRG) and posing its extension to n -point scattering as an open exercise; the analogous n -photon question is not itself part of Q44 but is the natural companion problem, and is the one this note addresses.

This note reports our results on the spinning $n = 5$ (photon) case.

Remark 1.1 (Scope). *This note makes no claim about the Classical Regge Growth conjecture at $n = 5$. That question — what the correct generalization of the classical Regge limit to $n \geq 5$ particles even is, and whether the CRG bound holds there —*

is treated separately, as an open problem with supporting (negative) computational evidence, in a companion note [4]. Nothing in the present note depends on Regge growth being true, false, or well-defined at $n = 5$.

Organization. Section 2 fixes conventions. Section 3 gives the S_5 -representation on Mandelstam invariants and its invariant ring, cross-checked against HLMM. Section 4 derives the photon polarization data structure at $n = 5$. Section 5 introduces the gauge-invariant building blocks, including the new “triple sandwich” atom. Section 6 proves a parity theorem. Section 7 gives the classification through mass dimension 11. Section 8 summarizes the verification methodology and documents errors caught in the process. Section 9 discusses scope and outlook.

2. SETUP AND CONVENTIONS

We work with $n = 5$ massless momenta $p_1, \dots, p_5$ in D -dimensional Minkowski space (mostly-plus signature), $p_i^2 = 0$, $\sum_i p_i = 0$, with D generic (large enough that no Gram-determinant relation among the p_i and polarizations ε_i is active — the same regime HLMM and CGGHJM use). Each photon carries a polarization vector ε_i subject to $\varepsilon_i \cdot p_i = 0$ and the residual gauge redundancy $\varepsilon_i \rightarrow \varepsilon_i + \zeta_i p_i$. We write $d_{ab} := p_a \cdot p_b$ and $s_{ab} := -2d_{ab}$.

Proposition 2.1 (Standard fact). *Suppose $S = V^\mu (\varepsilon_i)_\mu$ is a Lorentz scalar, linear in a single ε_i (with V built from any other data — other momenta, other polarizations, and possibly other invariant tensors such as the Levi-Civita symbol) and invariant under $\varepsilon_i \rightarrow \varepsilon_i + \zeta_i p_i$ for all ζ_i . If there exists a vector n , not built from ε_i , with $p_i \cdot n \neq 0$ (guaranteed here by the genericity of the kinematics fixed above), then S depends on ε_i only through the field strength $F_{\mu\nu}^i := p_{i,\mu} \varepsilon_{i,\nu} - p_{i,\nu} \varepsilon_{i,\mu}$: explicitly, $S = F_{\mu\nu}^i n^\mu V^\nu / (p_i \cdot n)$.*

Proof. Gauge invariance forces $p_i \cdot V = 0$ (from $S \rightarrow S + \zeta_i p_i \cdot V$). Then, using $p_i \cdot V = 0$,

$$F_{\mu\nu}^i n^\mu V^\nu = (p_i \cdot n)(\varepsilon_i \cdot V) - (\varepsilon_i \cdot n)(p_i \cdot V) = (p_i \cdot n) S,$$

which rearranges to the claimed identity. Note that this holds regardless of what other invariant tensors were used to build V ; only linearity in ε_i is used, which is why the claim does *not* extend to atoms with degree ≥ 2 dependence on a single ε_i (e.g. a symmetric form $\varepsilon_i^\mu Q_{\mu\nu} \varepsilon_i^\nu$ with $Q_{\mu\nu} p_i^\nu = 0$ is gauge invariant but generically not expressible through F^i alone). Every atom used below is linear in each polarization it depends on, so this restriction is never an issue in what follows. $\square$

We build every invariant below out of Lorentz contractions of the F^i and the momenta.

3. THE S_5 -REPRESENTATION ON MANDELSTAM INVARIANTS, AND ITS INVARIANT RING

3.1. Representation. The $\binom{5}{2} = 10$ “raw” invariants s_{ij} carry the permutation representation of S_5 on the edges of the complete graph K_5 . Momentum conservation gives, for each particle k , the constraint $\sum_{j \neq k} s_{kj} = 0$ — the edge-to-vertex incidence map of K_5 , which is S_5 -equivariant. The space of independent Mandelstam invariants is the *kernel* of this map: automatically an S_5 -representation, of dimension $10 - 5 = 5$.

Theorem 3.1. *The 5-dimensional representation of S_5 described above is irreducible.*

Proof. The permutation module on (unordered) pairs of a 5-element set decomposes, by the classical Johnson-scheme (Gelfand pair) decomposition for S_5 acting on 2-subsets, as $M^{(3,2)} \cong V^{[5]} \oplus V^{[4,1]} \oplus V^{[3,2]}$, of dimensions $1+4+5 = 10$; the vertex permutation module is $M^{(4,1)} \cong V^{[5]} \oplus V^{[4,1]}$, of dimension 5. The edge-to-vertex incidence map is S_5 -equivariant and, by the dimension count above, surjective with 5-dimensional kernel; since its image must be an S_5 -subrepresentation of $M^{(4,1)}$ of the full dimension 5, the image is all of $V^{[5]} \oplus V^{[4,1]}$, and the kernel is exactly the complementary irreducible summand $V^{[3,2]}$. This is irreducible by construction. $\square$

Remark 3.2 (Independent numerical cross-check). *We additionally verified this by direct computation: the character norm $\|\chi\|^2 = \frac{1}{|S_5|} \sum_g |\chi(g)|^2$, computed from the hand-derived formula $\chi(g) = \binom{\text{fix}(g)}{2} + c_2(g) - \text{fix}(g)$ (where $\text{fix}(g)$ is the number of fixed points and $c_2(g)$ the number of 2-cycles of g — exactly the character of $M^{(3,2)}$ minus that of $M^{(4,1)}$), equals exactly 1 when checked against the trace of the explicit representation matrices on all 7 conjugacy classes of S_5 (values 5, 1, 1, -1 , 1, -1 , 0 on classes $1^5, 21^3, 2^21, 31^2, 32, 41, 5$, matching the standard character table row for the partition $[3, 2]$). A brute-force homomorphism test, $\rho(g_1)\rho(g_2) = \rho(g_1g_2)$ on 40 random pairs, found zero failures.*

This is qualitatively different from $n = 4$, where the analogous 2-dimensional space (s, t) is *also* irreducible (as a representation of S_3), but is realized by literal permutation of (s, t, u) among themselves — no such simple description exists for $n = 5$.

3.2. The invariant ring is not free. For $n = 4$, S_3 acts on (s, t) as a genuine reflection group, so by the Chevalley–Shephard–Todd theorem the invariant ring is a free polynomial ring (generators in degrees 2 and 3).

Theorem 3.3. *The ring of S_5 -invariant polynomials on the 5-dimensional representation of Theorem 3.1 is not a free polynomial ring.*

Proof. By Chevalley–Shephard–Todd, the invariant ring is free if and only if S_5 acts on this representation as a group generated by pseudo-reflections (elements with eigenvalue 1 of multiplicity exactly 4). A pseudo-reflection $g \neq e$ on this 5-dimensional representation would have eigenvalues $(1, 1, 1, 1, \lambda)$ with $\lambda \neq 1$ a root of unity, hence $\chi(g) = 4 + \lambda$; since every character value of S_5 is a rational integer, $4 + \lambda \in \mathbb{Z}$ forces $\lambda = -1$ (the only non-trivial real root of unity), i.e. $\chi(g) = 3$ is the unique character value a pseudo-reflection could have. The character values computed in Theorem 3.1 on the 7 conjugacy classes $(5, 1, 1, -1, 1, -1, 0)$ never equal 3 for any non-identity class, so *no* element of S_5 can be a pseudo-reflection in this representation. Hence the invariant ring is not free. $\square$

Remark 3.4 (Independent numerical cross-check). *This was also checked directly: the exact eigenvalues (computed symbolically, no numerical approximation) of one representative of each of the 7 conjugacy classes of S_5 show that no non-identity element has eigenvalue 1 with multiplicity 4 — even transpositions, the closest candidates, fix only a 3-dimensional subspace ($\chi = 1 = a - b$ with $a + b = 5$ forces $a = 3$), matching the general argument above.*

3.3. Complete ring structure. We computed the complete ring structure two independent ways.

Molien series (character theory) gives Hilbert-series coefficients

$$1, 0, 1, 1, 2, 2, 5, 4, 8, 9, 13, 15$$

at degrees 0 through 11.

Direct computation via Singular’s `finvar.lib` (Noether normalization together with Kemper–Steel secondary invariants) on the same explicit S_5 generator matrices gives 5 primary invariants of degrees 2, 3, 4, 5, 6, and 6 secondary invariants of degrees 0, 6, 7, 8, 9, 15 (of sizes 1, 158, 283, 495, 638, 3565 terms respectively). The reconstructed Hilbert series

$$H(x) = \frac{1 + x^6 + x^7 + x^8 + x^9 + x^{15}}{(1 - x^2)(1 - x^3)(1 - x^4)(1 - x^5)(1 - x^6)}$$

matches the independent Molien series exactly in all 12 degrees checked.

Remark 3.5 (Validation against the literature). *This scalar-case ring structure is already published: HLMM [1], eq. (5.50a), give exactly the same Hilbert series, the same primary degrees (2, 3, 4, 5, 6), and the same secondary degrees (0, 6, 7, 8, 9, 15), for the identical problem (five identical massless scalars, generic D), using the same computational tool. We report this explicitly as a validation checkpoint confirming our machinery before extending it to the spinning case, which is where the genuinely new content of this note begins; the scalar ring itself is not claimed as a new result.*

4. PHOTON POLARIZATION DATA AT $n = 5$

4.1. Dimensional count. For $n = 4$, the four (generic) momenta span a 3-dimensional “scattering plane”; each polarization splits into a transverse part $\varepsilon_i^\perp$ (living in the $(D-3)$ -dimensional complement, acted on by the little group $\text{SO}(D-3)$) and an in-plane part $\varepsilon_i^\parallel$. Imposing $\varepsilon_i \cdot p_i = 0$ and residual gauge freedom reduces $\varepsilon_i^\parallel$ to a *single* complex parameter α_i per particle ([2], eq. 2.11).

Repeating this count for $n = 5$: the five momenta span a 4-dimensional scattering plane; the transverse little group is $\text{SO}(D-4)$, and — after the same two reductions — $\varepsilon_i^\parallel$ is left with *two* independent physical parameters (α_i, α'_i) per particle.

4.2. Explicit formula and its inverse. Writing $w_{jk} := p_j/s_{1j} - p_k/s_{1k}$, we derive

$$\varepsilon_1^\parallel = \alpha_1 \cdot w_{23} + \alpha'_1 \cdot w_{24} + a_1 \cdot p_1, \quad a_1 \text{ pure gauge.}$$

Verification includes: the orthogonality identity $p_1 \cdot w_{jk} = 0$ for all 6 pairs; an explicit rank computation (exact, symbolic, treating the $p_i \cdot p_j$ as the 5 free Mandelstam variables) showing the “raw” span of the w_{jk} (before quotienting by the gauge direction p_1) has rank exactly 3, with p_1 itself contained in that span (giving physical dimension $3 - 1 = 2$) — a naive first count had incorrectly predicted rank 2, an off-by-one error caught by comparing against the known $n = 4$ case, where the analogous raw span also has rank one more than the physical dimension. We also checked, by explicit linear solve, that the omitted pairs $w_{25}, w_{34}, w_{35}, w_{45}$ are each an exact linear combination of the chosen basis $\{w_{23}, w_{24}, p_1\}$, confirming no information is lost or double-counted.

The inverse relation, expressing (α_1, α'_1) in terms of the field-strength contractions $C_{jk} := p_j \cdot F^1 \cdot p_k$, is a linear 2×2 system whose matrix is a *non-diagonal*

function of the Mandelstam invariants — genuinely coupled, unlike the trivial 1×1 case at $n = 4$.

Remark 4.1 (Avoiding a spurious pivot dependence). *An initial attempt to fix a single “pivot” particle and express all polarization data in terms of (α_i, α'_i) explicitly ran into a real obstruction: transition functions between different pivot choices are not always trivializable by rescaling, and the construction gives a vector bundle of rank 2 over the choice of pivot, not a linear representation of S_4 . We do not use that construction in what follows; every result from Section 5 onward is built from manifestly pivot-independent, gauge-invariant contractions of the field strengths F^i directly, which sidesteps this issue entirely.*

5. GAUGE-INVARIANT BUILDING BLOCKS (“ATOMS”)

We build every multilinear invariant from four families of manifestly gauge-invariant contractions of the field strengths $F_{\mu\nu}^i$ and momenta, each verified numerically with explicit $D = 6$ kinematics (gauge shifts applied to check invariance to machine precision).

atom	definition	pol. slots	mom. degree
$F^i:F^j$	$F_{\mu\nu}^i F^{j,\mu\nu}$	2	2
C_{ab}^k	$p_a \cdot F^k \cdot p_b$	1	3
M_{ab}^{ij}	$p_a \cdot F^i \cdot F^j \cdot p_b$ (“double sandwich”)	2	4
T_{ab}^{ijk}	$p_a \cdot F^i \cdot F^j \cdot F^k \cdot p_b$ (“triple sandwich”, new)	3	5

$F^i:F^j$ admits a closed form in terms of (α, α') :

$$F^1:F^2 = 2(p_1 \cdot p_2)(\varepsilon_1^\perp \cdot \varepsilon_2^\perp) + F^1:F^2|_{\parallel},$$

$$F^1:F^2|_{\parallel} = -(\alpha_1\alpha_2 + \alpha'_1\alpha'_2) - Q \left[\frac{\alpha_1\alpha'_2}{2d_{13}d_{24}} + \frac{\alpha'_1\alpha_2}{2d_{14}d_{23}} \right],$$

$$Q = d_{12}^2 + d_{12}d_{13} + d_{12}d_{14} + d_{12}d_{23} + d_{12}d_{24} + d_{13}d_{24} + d_{14}d_{23}.$$

Notably, the coefficients of $\alpha_1\alpha_2$ and $\alpha'_1\alpha'_2$ are exactly -1 with no kinematic dependence; only the cross terms carry the Q -dependence. The same Q appears in the inverse polarization formula of §4 and in the linear relations between the six C_{jk} for a fixed particle — a consistency thread across independently-derived pieces.

M_{ab}^{ij} and T_{ab}^{ijk} are built by propagating a momentum through 2 (resp. 3) field strengths in sequence,

$$v \mapsto v \cdot F^m := (v \cdot p_m)\varepsilon_m - (v \cdot \varepsilon_m)p_m,$$

representing intermediate vectors purely by their dot products with all p_x, ε_x (never explicit components), which avoids any choice of basis. T is new to this work; the propagation chain was validated by checking that applying it twice reproduces the independently verified M , before trusting a third application.

6. A PARITY THEOREM

Theorem 6.1. *Every multilinear five-photon invariant (linear in each of the 5 polarizations) built using only the metric $\eta^{\mu\nu}$ to perform Lorentz contractions (no Levi-Civita tensor; cf. Remark 7.1) has odd total momentum degree.*

Proof. By Proposition 2.1, every polarization enters through its field strength F^i . Consider the complete contraction graph of the invariant: all 5 field strengths $F^1, \dots, F^5$, with their 10 free indices in total, each saturated by contracting with either another field-strength index (an “ F - F ” contraction, using 2 indices and 0 extra momenta) or an external momentum (an “ F -momentum” contraction, using 1 index and 1 extra momentum) — these are the only two options when contractions use only the metric. (A self-contraction $\eta^{\mu\nu} F_{\mu\nu}^i$ of a single field strength with itself vanishes identically by antisymmetry of F^i , so every nontrivial F - F contraction pairs two distinct field strengths F^i, F^j , $i \neq j$; this does not affect the count below.) If there are x F - F contractions and y F -momentum contractions overall, saturating all 10 indices gives $2x + y = 10$, so y is even. The total momentum degree of the invariant is $5 + y$ (one intrinsic momentum per field strength, plus y external contractions), possibly times a Mandelstam multiplier of even degree (built from $d_{ab} = p_a \cdot p_b$); since y is even, the total degree is $\equiv 5 \equiv 1 \pmod{2}$ regardless of the contraction pattern or multiplier used. Note this argument is applied directly to the invariant’s full contraction graph, whether or not it factors as a product of independently-contracted pieces (each of the four atoms of Section 5 is one such factor, but the argument does not require this factorization). $\square$

Remark 6.2 (Sanity check). *The identical argument for $n = 4$ gives total degree $\equiv 0 \pmod{2}$, consistent with all known $n = 4$ operators (e.g. $(\varepsilon_i \cdot \varepsilon_j)(\varepsilon_k \cdot \varepsilon_l)$ -type) having even degree.*

Corollary 6.3. *Mass dimensions 8 and 10 are identically zero for $n = 5$.*

7. CLASSIFICATION THROUGH MASS DIMENSION 11

Remark 7.1 (Scope of the classification). *The classification below is complete within the family of invariants that factor as a product of the four atoms of Section 5, each depending on a disjoint subset of polarizations, contracted using only the metric (no Levi-Civita tensor, which we have not explored). Two distinct restrictions are implicit here, and we flag both explicitly rather than letting the classification’s scope be inferred: (i) we do not claim the four listed atoms are the only metric-only gauge-invariant building blocks — e.g. a single connected contraction graph touching all 5 field strengths at once (such as a cyclic trace $F^1_{\mu}{}^{\nu} F^2_{\nu}{}^{\rho} \dots F^5_{\rho}{}^{\mu}$) is not obviously reducible to a product of the four atoms above, and we have not exhaustively ruled out such “entangled” topologies; (ii) we do not claim completeness once Levi-Civita contractions are allowed. Neither gap affects Theorem 6.1 (whose proof applies directly to the full contraction graph of any metric-only invariant, factored or not), but both are real boundaries of the degree-by-degree dimension count in this section, stated here rather than left implicit.*

degree	dimension	status
7	0	proven: any candidate vanishes identically under S_5 -symmetrization
8	0	Corollary 6.3
9	3	exact; combinatorially complete (below)
10	0	Corollary 6.3
11	16	exact; combinatorially complete (below)

Degree 7. Any degree-7 candidate in the known topologies vanishes identically under S_5 -symmetrization, by an antisymmetry mechanism: the singlet C_{ab}^k is antisymmetric in a, b , and some permutation in the S_5 orbit always swaps that pair, cancelling the sum.

Degree 9 (dimension 3). The only slot-partitions of the 5 polarizations reachable at degree 9 with the atoms of Section 5 are $F:F + F:F + C$ (base degree 7, needing a degree-2 Mandelstam corrector) and $F:F + M + C$ (degree $2 + 4 + 3 = 9$ exactly, no corrector) — $M + M + C$ is degree 11, too high, and T 's minimum degree is 11 ($T + C + C = 5 + 3 + 3$). An exact-arithmetic rank/nullspace computation (exact rational matrices, cross-checked against a 60-digit arbitrary-precision computation on independent kinematic samples) confirms dimension exactly 3, with no further direction found after exhausting all contraction variants of both partitions.

Degree 11 (dimension 16). The analogous partition-counting argument gives four topologies — $F:F + F:F + C$ (with a degree-4 corrector), $F:F + 3 \times C$ (exact), $F:F + M + C$ (with a degree-2 corrector), $M + M + C$ (exact) — which a first pass showed span a 15-dimensional space. An exhaustive combinatorial sweep over all contraction variants within these four topologies (887 seeds, two independent batches of 20 random high-precision kinematic samples each, 60-digit precision, residuals $\sim 10^{-61}$) confirmed rank exactly 15, ruling out an incomplete-seed-set artifact.

A fifth topology, $T + C + C$ (the new triple-sandwich atom, exact degree $5 + 3 + 3 = 11$), was identified by the same slot/degree counting argument — the *only* new partition the T atom opens up at this degree — and shown to add exactly *one* further dimension, confirmed in two independent batches of high-precision kinematic samples, and shown not to grow further after also exhausting the other two partitions T opens at degree 11 with a corrector ($T + F:F$, $T + M$) — 1297 total seeds tested, rank stable at 16.

8. VERIFICATION METHODOLOGY

Every numeric result above was independently re-derived by at least one alternative method before being reported as final, and the process caught several real errors, corrected in place:

- A convention mismatch (σ versus σ^{-1} , i.e. an accidental anti-homomorphism instead of a genuine group homomorphism) in an early derivation of the S_5 action, caught by a full generator-set change-of-basis test (trace/determinant matching alone is insufficient, since both are constant on conjugacy classes and do not distinguish an anti-homomorphism from a homomorphism).
- A silent numerical corruption from forcing exact rational coefficients to nearby integers, caught by comparing against exact symbolic coefficient matching instead.
- A homogeneity bug (a hand-typed seed of the wrong degree contaminating a rank computation), caught by adding a permanent degree-homogeneity guardrail to the rank-computation utility.
- A floating-point precision failure at degree eleven and above (spurious, sample-dependent rank estimates in double precision), diagnosed and resolved by switching to arbitrary-precision or exact rational arithmetic.

We regard this track record as informative about the reliability of the numbers reported here, but it does not substitute for independent external verification, which we consider the necessary next step before wider circulation of this note.

9. DISCUSSION AND OUTLOOK

This note establishes the S_5 -representation-theoretic and combinatorial foundation for gauge-invariant five-photon operators through mass dimension 11, and reports a new building block (the triple field-strength sandwich T) that genuinely extends the operator basis at degree 11 beyond what the previously known atoms capture. All of this is independent of, and does not bear on, whether or how the Classical Regge Growth conjecture generalizes to $n \geq 5$; we discuss that question, together with supporting (negative) computational evidence, in a companion note [4].

Natural continuations: (i) settle whether a fifth atom type beyond Section 5 exists, either via Levi-Civita contractions or via “entangled” metric-only topologies not factoring into the four listed atoms (both flagged in Remark 7.1), which would upgrade Section 7 from a classification within a stated family to a genuine completeness theorem; (ii) extend the classification to degree 13; (iii) investigate whether the ring structure found here is Cohen–Macaulay, addressing directly the question HLMM leave open for spinning operators at $n \geq 5$.

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